

5.6.40. A set of blocks contains blocks of heights 1, 2, and 4 centimeters. Imagine constructing tower by piling blocks of different heights directly on top of one another. (A tower of height 6 cm could be obtained using six 1-cm blocks, three 2-cm blocks one 2-cm block with one 4-cm block on top, one 4-cm block with one 2-cm block on top, and so forth.) Let t_n be the number of ways to construct a tower of height n cm using blocks from the set. (Assume an unlimited supply of blocks of each size.) Find a recurrence relation for $t_1, t_2, t_3, \dots$. Justify your answer.

Suppose t_n is the number of ways to construct a tower of height n .

Let $T = \{1, 2, 4\}$

The set T contains all different types of blocks, defined by their height in centimeters.

Before finding the recurrence relation, we should try to find the initial values of the sequence. Since our recurrence will depend on previous terms, these initial values will give us a starting point from which the recurrence can be applied.

For the initial values:

$$t_1 : (1)$$

$$t_1 = 1$$

$$t_2 : (2), (1, 1)$$

$$t_2 = 2$$

$$t_3 : (1, 2), (1, 1, 1), (2, 1)$$

$$t_3 = 3$$

$$t_4 : (4), (1, 1, 1, 1), (2, 2), (1, 1, 2), (1, 2, 1), (2, 1, 1)$$

$$t_4 = 6$$

Now that we have a set of initial values within reason, we can try to find a recurrence relation that describes how the remaining terms can be obtained from previous term.

Consider a tower of height n . The block on top of the tower must have a height contained in our set T .

If the top block has height 1 cm, then the remaining tower must have height $n - 1$, which means there're t_{n-1} ways to construct this tower. The same thing can be said for 2 and 4 cm respectively.

Therefore, the number of ways to construct a tower of height n is:

$$t_n = t_{n-1} + t_{n-2} + t_{n-4}, \text{ for } n \geq 5$$

Since the smallest subscript starts at t_1 , n cannot be less than 5. For example, if $n = 4$:

$$t_4 = t_3 + t_2 + t_0$$

But t_0 is not defined. However, if $n = 5$:

$$t_5 = t_4 + t_3 + t_1$$

Therefore, the recurrence relation applies for $n \geq 5$.

Solution: $t_n = t_{n-1} + t_{n-2} + t_{n-4}$, for $n \geq 5$, and $t_1 = 1$, $t_2 = 2$,
 $t_3 = 3$, and $t_4 = 6$